

HEALTHCARE DATA ANALYTICS USING POWER BI

PROJECT REPORT

Submitted By:
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Course:
Data Analytics

Tools Used:
Power BI Desktop | Power Query | DAX

Project Title:
Healthcare Data Analytics Dashboard

1. Project Objective

The objective of this project is to analyze healthcare data using Power BI and identify insights related to patient demographics, health conditions, hospital visits and treatment costs. An interactive dashboard was created to help stakeholders understand the healthcare data.

2. Dataset Description

The dataset contains 500 patient records and 10 columns.

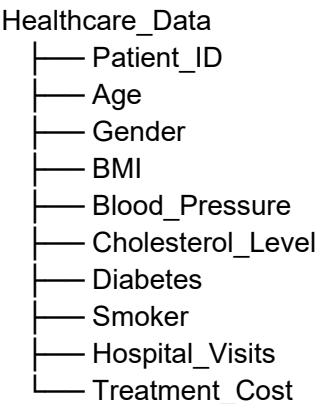
Column	Description
Patient_ID	Unique patient identifier
Age	Age of patient
Gender	Gender of patient
BMI	Body Mass Index
Blood_Pressure	Patient blood pressure
Cholesterol_Level	Cholesterol level
Diabetes	Diabetes status
Smoker	Smoking status
Hospital_Visits	Number of hospital visits
Treatment_Cost	Cost of treatment

3. Data Preparation

The healthcare dataset was imported into Power BI and cleaned using Power Query. Column names and data types were checked, missing values were reviewed, and the final table was renamed to Healthcare_Data. The cleaned data was then loaded into Power BI.

4. Data Model

The project contains one table named Healthcare_Data. Since there is only one table, no relationship between tables is required.



5. DAX Calculations

Average Treatment Cost

```
Average Treatment Cost =  
AVERAGE(Healthcare_Data[Treatment_Cost])
```

Purpose: Calculates the average treatment cost.

Average BMI

```
Average BMI =  
AVERAGE(Healthcare_Data[BMI])
```

Purpose: Calculates the average BMI.

Total Hospital Visits

```
Total Hospital Visits =  
SUM(Healthcare_Data[Hospital_Visits])
```

Purpose: Calculates the total number of hospital visits.

Age Group

```
Age Group =  
SWITCH(  
    TRUE(),  
    Healthcare_Data[Age] <= 30, "18-30",  
    Healthcare_Data[Age] <= 45, "31-45",  
    Healthcare_Data[Age] <= 60, "46-60",  
    "61+"  
)
```

Purpose: Groups patients by age for analysis.

6. Dashboard Components

- Average Treatment Cost – KPI
- Average BMI – KPI
- Total Hospital Visits – KPI
- Age Distribution
- Gender Distribution
- BMI vs Treatment Cost
- Average Hospital Visits by Age Group
- Smoking Status
- Diabetes Status
- Gender, Diabetes and Smoking filters

HEALTHCARE DATA ANALYTICS DASHBOARD

Average Treatment Cost

\$10.73K

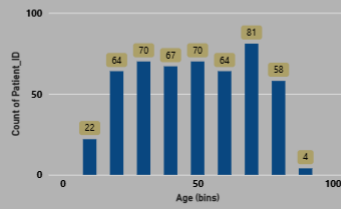
Average BMI

27.62

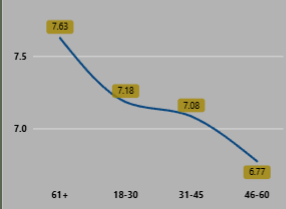
Total Hospital Visits

4K

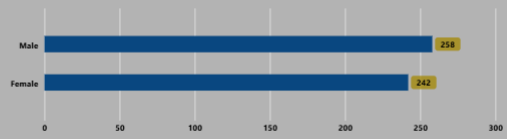
Age Distribution of Patients



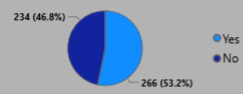
Average Hospital Visits by Age Group



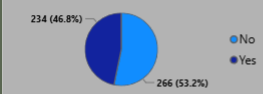
Gender Distribution of Patients



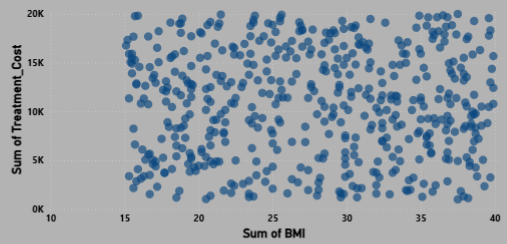
Smoking Status



Diabetes Status



BMI vs Treatment Cost



Gender Filter

- ☐ Female
☐ Male

Diabetes Filter

- ☐ No
☐ Yes

Smoking Filter

- ☐ No
☐ Yes

7. Key Insights

Hospital Visits by Age Group

The 61+ age group has the highest average hospital visits, at approximately 7.63 visits.

Diabetes and Treatment Cost

Non-diabetic patients have an average treatment cost of approximately 11,021.85, compared with 10,401.32 for diabetic patients.

BMI and Treatment Cost

BMI and treatment cost show virtually no linear correlation in this dataset.

Gender Distribution

The dataset contains 258 male patients and 242 female patients.

Smoking Status

Average hospital visits are very similar for smokers and non-smokers. Non-smokers average approximately 7.28 visits compared with 7.24 for smokers. Non-smokers also have a slightly higher average treatment cost.

8. Conclusion

The Healthcare Data Analytics project demonstrates how Power BI can be used to analyze patient demographics, health conditions, hospital visits and treatment costs. The interactive dashboard allows users to filter the analysis by gender, diabetes and smoking status and provides a clear visual summary of the dataset.

9. Tools & Technologies

- Power BI Desktop
- Power Query
- DAX
- CSV Dataset
- Data Visualization

10. Project Deliverables

- Power BI (.pbix) file
- PDF project report
- Dashboard screenshot
- Data model documentation
- DAX documentation
- Key insights and interpretation